

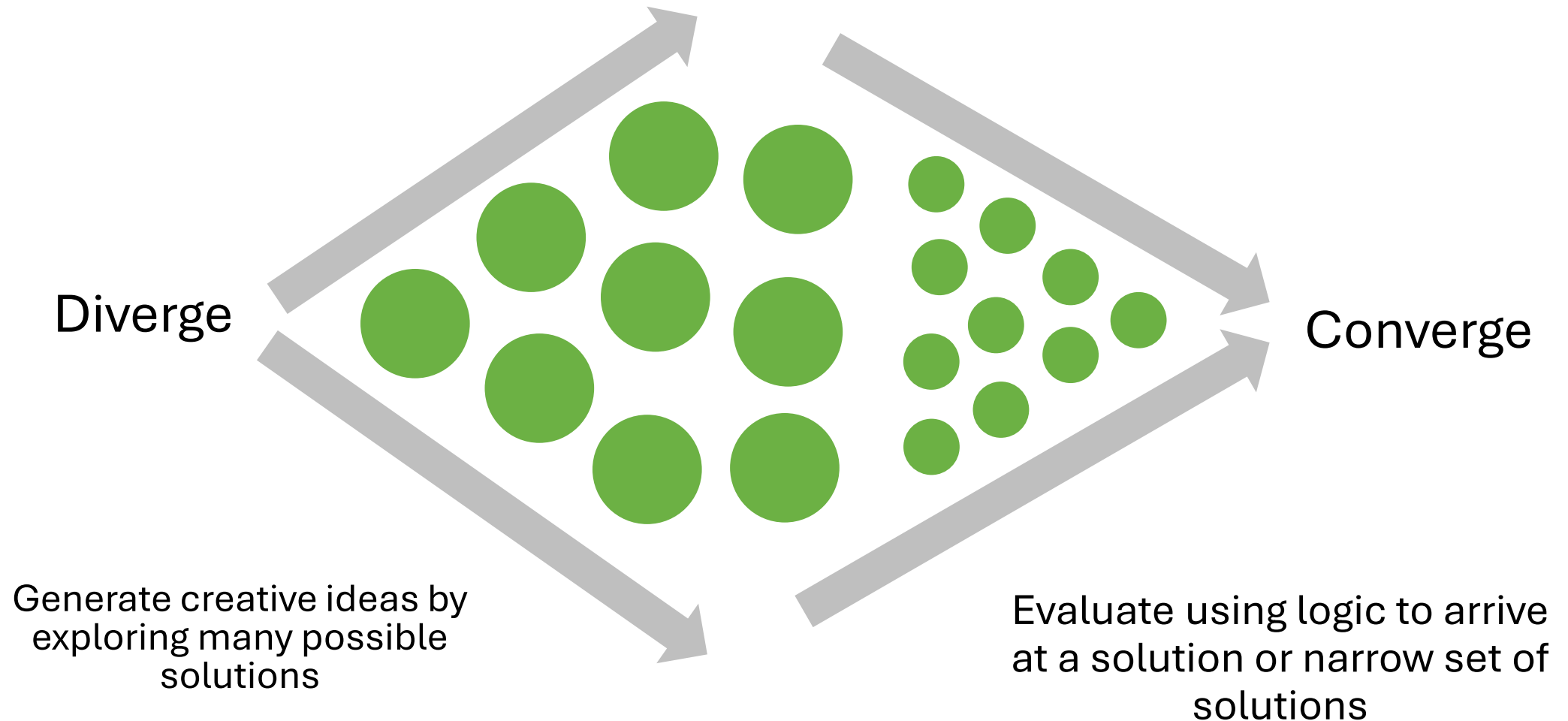
Tools for a Shared Understanding

Visualizing Requirements with
Journey Maps,
Lean Canvas Model, and
Sequence Diagrams

Intro / Recap

- School solution to the Journey Map of Market to Market
- Diverge / converge tools

Divergent vs. Convergent Thinking



Diverge

- **Interviews**
Learn what people say and why they think it matters.
- **Mash-ups**
Combine patterns from different domains to imagine new approaches without designing solutions yet.
- **Journey Maps**
Expose flow, friction, handoffs, and emotional cost across time.
- **Problem Framing**
Decide what problem you are actually trying to solve.
- **Assumption Mapping**
Surface what must be true for your idea to work.

Converge

- **Synthesis / Affinity Mapping**
Turn raw input into shared understanding.
- **Opportunity Prioritization**
Decide where effort is worth spending.
- **Tradeoff Analysis**
Make constraints explicit and choose deliberately.
- **Sequence Diagrams**
Make responsibilities, ordering, and interactions explicit so requirements can be validated.
- **Prototyping + Validation**
Test the riskiest assumptions before scaling.

Lean Model Canvas

Capture strategic business aspects to align solutions with needs

Before You Write Requirements

WHEN

- When the solution is clearer than the problem
- When stakeholders say “We need to build...”
- When you’re about to commit engineering time
- When success isn’t clearly defined

WHY

- Most projects fail upstream
- Wrong customer
- Weak problem
- No measurable value
- Requirements optimize the wrong thing

What It Forces You To Clarify

- Who exactly is this for?
- What painful problem are we solving?
- How are they solving it today?
- Why would they switch?
- What behavior proves this is working?

RELAYFLOW - LEAN MODEL CANVAS

Problem

- Uneven wave starts create bottlenecks
- Exchange zones get congested
- Volunteer coordination gaps
- Runner frustration from delays
- Inefficient van utilization
- Limited real-time visibility

Customer Segments

- Race directors (primary buyer)
- Event organizations
- Race volunteers
- Participating runners
- Event sponsors
- Timing companies

Unique Value Proposition

Optimize wave starts, exchange flow, and van routing in real time to reduce congestion and improve runner experience

Solution (Level)

- Real-time race coordination dashboard
- Dynamic wave rebalancing recommendations
- Exchange congestion heat mapping
- Van routing optimization
- Volunteer reporting interface
- Predictive bottleneck alerts

Channels

- Direct sales to race organizations
- Sports event management networks
- Endurance event associations
- Race timing company partnerships

Revenue Streams

- Per-event licensing fee
- Annual subscription for race series
- Tiered pricing by participant count
- Premium analytics add-ons

Existing Alternatives

- Clipboards and spreadsheets
- Radio communication systems
- Static wave planning
- Manual volunteer reporting

Cost Structure

- Software development
- Cloud hosting infrastructure
- On-site race support
- Customer onboarding

Key Metrics

- Average wait time at exchanges
- Wave variance deviation
- On-time leg completion rate
- Runner satisfaction scores
- Volunteer response time
- Revenue per event

Unfair Advantage

- Historical race data analytics
- Predictive congestion modeling
- Integrated race operations expertise
- Real-time optimization algorithms

Sequence Diagrams

Show interactions and system behavior over time

Sequence Diagrams - When

- Confusion or ambiguity on workflow and interactions
- Uncertainty on where issue might arise
- Requirements delivered verbally

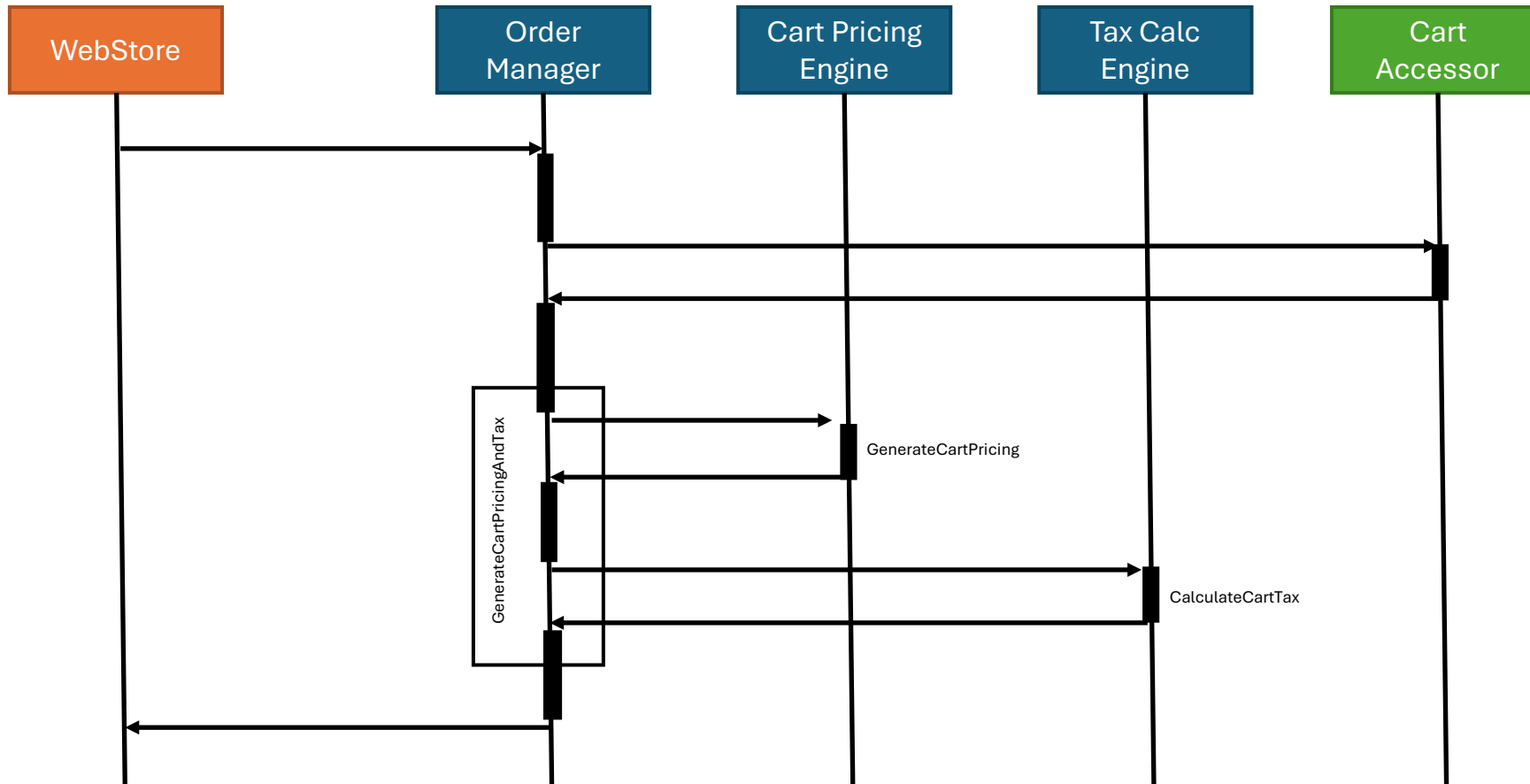
Sequence Diagrams - Why

- Understand complex actors, systems, and interactions
 - Changes between users / customers
 - Changes over time
- Clarify workflow
- Validate behavior and identify bottlenecks
- Pictures and diagrams are easier to understand and align than words

Sequence Diagrams - How

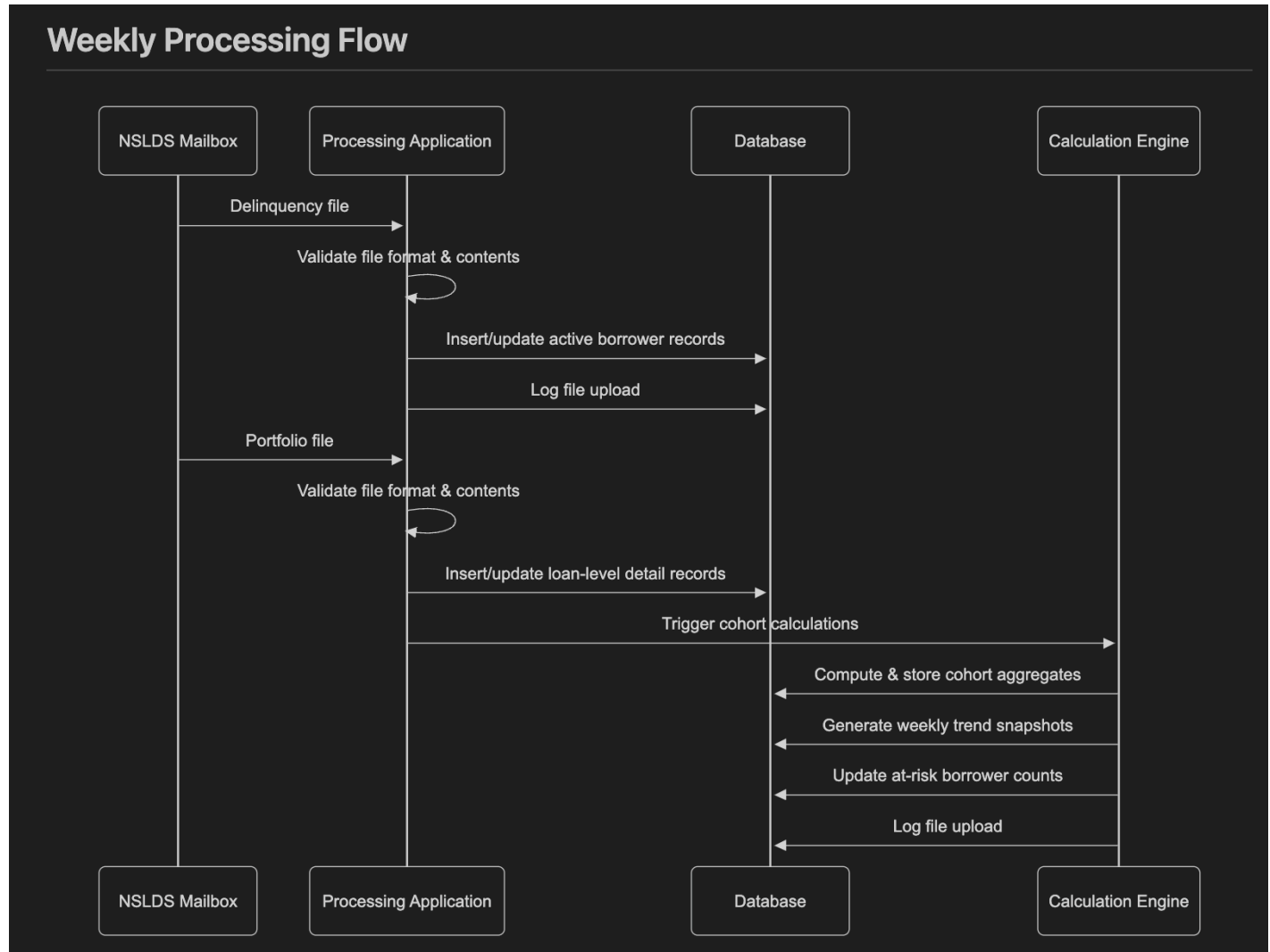
- Core Components
 - Actors (nouns),
 - Messages (verbs), and
 - Lifelines (time)
- Illustrate communication and control within the system

Sequence Diagram: Adding an item to the cart



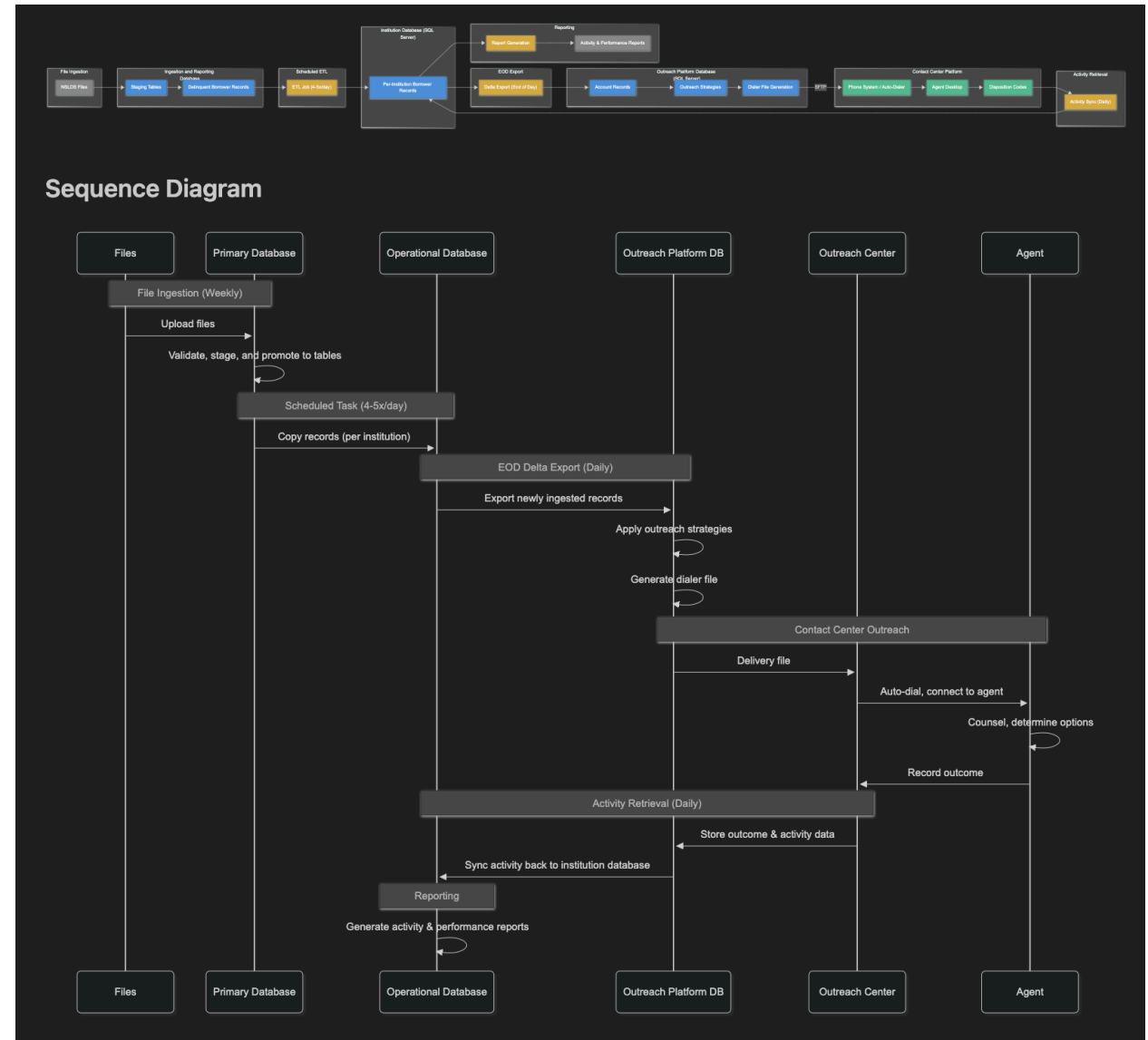
Sequence Diagram:

Recent Example





Sequence Diagram: Recent Example



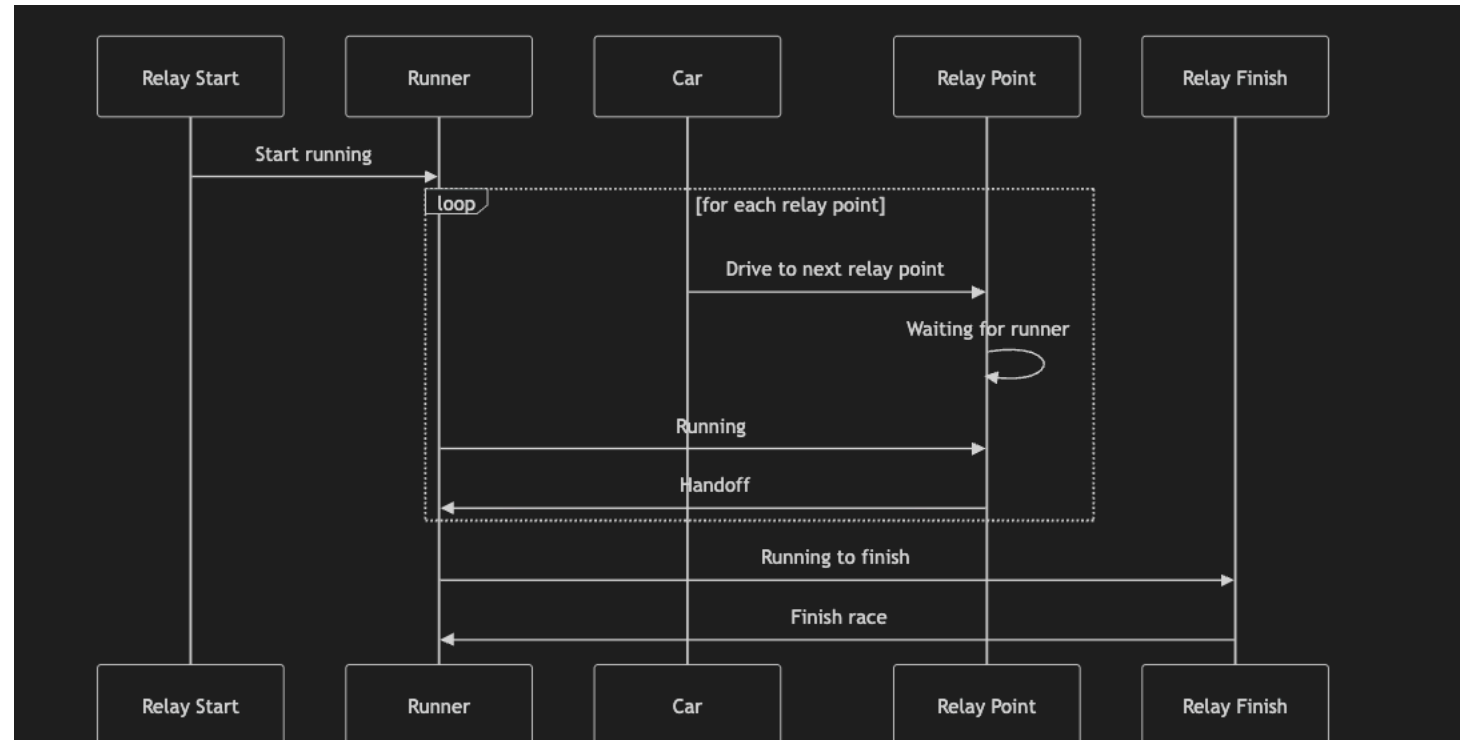
Market-to-Market Example (an antipattern)

```
``mermaid
sequenceDiagram
    participant Start as Relay Start
    participant Runner
    participant Car
    participant RelayPoint as Relay Point
    participant Finish as Relay Finish

    Start->>Runner: Start running

    loop for each relay point
        Car->>RelayPoint: Drive to next relay point
        RelayPoint->>RelayPoint: Waiting for runner
        Runner->>RelayPoint: Running
        RelayPoint ->> Runner: Handoff
    end

    Runner->>Finish: Running to finish
    Finish->>Runner: Finish race
    ...
```



Activity: Practice

Pick one of journey map, lean model canvas, or sequence diagram for a key interaction and try it for your project

Share time

Activity: Circle back

Go back to user stories - do you have any missing stories or missing details?